

Zoeb Mithaiwala

zoebmithaiwala1@gmail.com · +91 9033936249 · [linkedin.com/in/zoeb-mithaiwala-34447b146](https://www.linkedin.com/in/zoeb-mithaiwala-34447b146)

SUMMARY

ML infrastructure and systems engineer with 7+ years across Google and early-stage AI companies. Deep experience in GPU-accelerated video processing, low-latency inference optimization, and large-scale data pipelines. Proven track record shipping production ML systems — from CUDA kernels and TensorRT deployments to petabyte-scale content extraction. Excited by frontier research infrastructure where ML meets real-world perception and action.

EXPERIENCE

Ambient.ai

Aug 2024 – Present

Software Engineer — ML & Edge Infrastructure

GPU & Video Pipeline Optimization

- Integrated NVDEC with FFmpeg for hardware-accelerated video decoding, significantly reducing CPU overhead on edge appliances.
- Moved motion detection from CPU image processing to custom CUDA kernels, enabling real-time parallel frame analysis.
- Migrated VLM object detection models from PyTorch to TensorRT, achieving meaningful inference latency reductions on edge hardware.
- Optimized VLM preprocessing pipeline at multiple levels (batching, memory layout, stream pipelining) to reduce end-to-end inference latency.
- Optimized video encryption pipeline and shared-memory (SHM) buffer writes for WebRTC-based video streaming.

Edge Health & Observability

- Built edge appliance health notification system and compliance log generation tooling.
- Improved monitoring coverage for internal cloud services including Celery, GraphQL, and Django.

Google

Jun 2020 – Aug 2024

Software Engineer — Infrastructure & Data Pipelines

- Led a year-long project to extract structured content from PDFs, MS Office, and OpenOffice formats at scale — powering search indexing in Google Drive/Gmail and abuse/spam detection in Gmail.
- Built and migrated data processing infrastructure for sovereignty-compliant data localisation, handling 300K peak QPS and billions of queries per day across Google Docs, Sheets, Slides, Gmail, and Drive.
- Developed automated conversion fidelity measurement for Google Editors document imports using heuristic-based diffing across diverse document types.

Morgan Stanley

Aug 2019 – May 2020

Technology Associate — NLP & ML

- Built intent classification model for an internal chatbot; clustered seed questions by semantic similarity for training data curation.
- Developed answer discovery system from documents and keyword detection for audio transcripts using NLP and deep learning.

Morgan Stanley

May 2018 – Jul 2018

Software Engineering Intern

- Built sentiment detection models from audio and transcripts using NLP and deep learning techniques.

TECHNICAL SKILLS

Core Strengths: GPU programming (CUDA), inference optimization, video processing pipelines, large-scale data infra

ML / AI: TensorRT, PyTorch, VLMs, object detection, NLP, deep learning, FFmpeg/NVDEC

Languages: Python, C++, Java, JavaScript, TypeScript

Infrastructure: gRPC, Django, Flask, Spring, OpenSearch, Redis, Celery, Prometheus, Grafana, Kinesis

Cloud: AWS, GCP

EDUCATION

B.Tech in Computer Science & Engineering — NITK Surathkal | CGPA: 9.1 / 10.0